

SURVIVAL ANALYSIS WITH PYTHON

Survival analysis is a set of statistical approaches used to determine the time it takes for an event of interest to occur. We use survival analysis to study the survival **time** of terminally ill patients in hospice. Time here is measured in days.

Using this, we have analysed:

- which age group is vulnerable to risk
- which gender has significant effect on their survival time
- the median number of days of survival for patients.
- Factor affecting has more impact on patients' survival.

The following methods of survival analysis are performed:

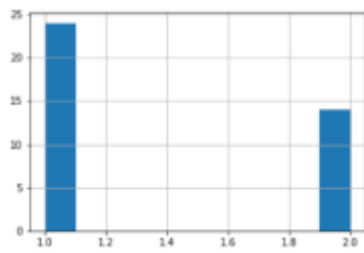
1) Kaplan-Meier plots to visualize survival curves.

2) Nelson-Aalen plots to visualize the cumulative hazard.

3) Log-Rank test to compare the survival curves of two or more groups

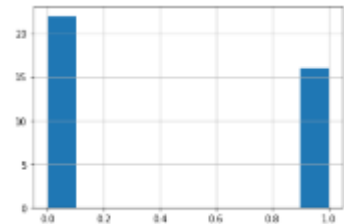
4) Cox-proportional hazards regression finds out the effect of different variables like age, sex, and weight on survival.

1. Kaplan Meiner plots:



The histogram of sex distribution where 1 (on the x-axis) refers to number of males and 2 (on the x-axis) refers to number of females.

The histogram of sex distribution where 0 (on the x-axis) refers to number of dead patients and 1 (on the x-axis) refers to number of patients who withdrew from the program.



Event Table of Survival Function:

One of the most crucial methods of this plot is the “event_table.” It gives us various information for our survival analysis.

The following terms are used:

event_at: It stores the value of survival days for the subjects.

at_risk: Total number of patients

censored: if the person is still alive, we will add him/her to the censored category.

observed: the number of subjects that died

Censored = Person who withdrew from the program

Observed = Persons who died.

Removed = Observed + Censored

Event_at	Removed	Observed	Censored	Entrance	At_risk
0.0	0	0	0	38	38
1.0	1	1	0	0	38
2.0	6	4	2	0	37
3.0	3	1	2	0	31
4.0	4	1	3	0	28
5.0	4	2	2	0	24
6.0	1	0	1	0	20
7.0	1	1	0	0	19
8.0	2	1	1	0	18
9.0	3	0	3	0	16
10.0	1	0	1	0	13
11.0	2	1	1	0	12
13.0	1	0	1	0	10
14.0	2	2	0	0	9
15.0	1	0	1	0	7
17.0	2	0	2	0	6
19.0	1	0	1	0	4
23.0	1	0	1	0	3
24.0	1	1	0	0	2
28.0	1	1	0	0	1

Let's divide our data into 2 groups: Male and Female. Our goal here is to check if there is any significant difference in survival rate if we divide our data set based on sex.

Event table for male group:

Event_at	Removed	Observed	Censored	Entrance	At_risk
0.0	0	0	0	24	24
1.0	1	1	0	0	24
2.0	5	4	1	0	23
3.0	3	1	2	0	18
5.0	2	1	1	0	15
6.0	1	0	1	0	13
8.0	2	1	1	0	12
9.0	1	0	1	0	10
10.0	1	0	1	0	9
11.0	1	1	0	0	8
13.0	1	0	1	0	7
14.0	2	2	0	0	6
15.0	1	0	1	0	4
17.0	1	0	1	0	3
19.0	1	0	1	0	2
28.0	1	1	0	0	1

Event table for female group:

Event_at	Removed	Observed	Censored	Entrance	At_risk
0.0	0	0	0	14	14
2.0	1	0	1	0	14
4.0	4	1	3	0	13
5.0	2	1	1	0	9
7.0	1	1	0	0	7
9.0	2	0	2	0	6
11.0	1	0	1	0	4
17.0	1	0	1	0	3
23.0	1	0	1	0	2
24.0	1	1	0	0	1

Timeline represents the number of survival days for each patient present in the dataset.

Prediction of the survival probability for the complete timeline (KM_estimate_survival):

It is observed that the Kaplan-Meier (survival probability) is decreasing as the timeline increases. Its values are depicted in KM_estimate_survival column. The plot is depicted below the table.

The median time patient's survived is calculated to be 14.

Prediction of the death probability for the complete timeline (KM_estimate_death):

Cumulative Density Function is used to determine the probability of death. The Cumulative density plot is depicted below.

Median time of Survival:

The amount of time remaining from the median survival time.

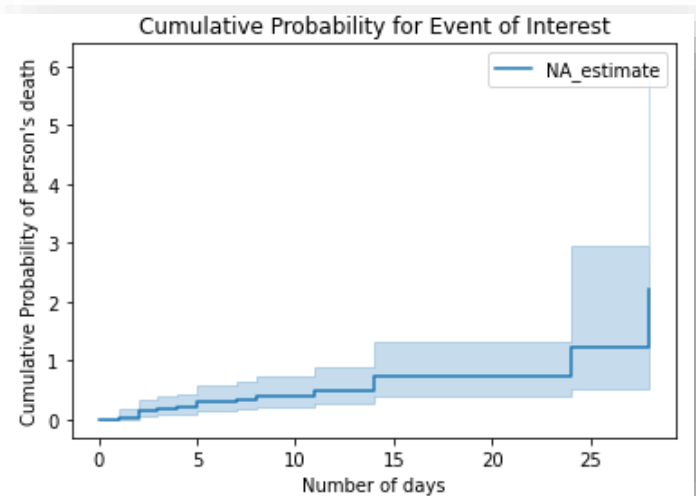
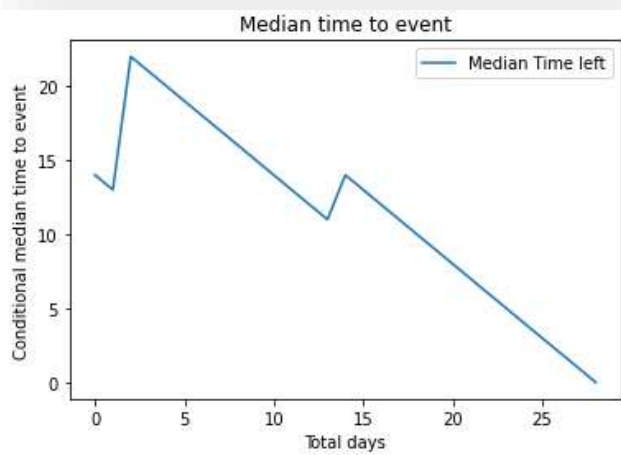
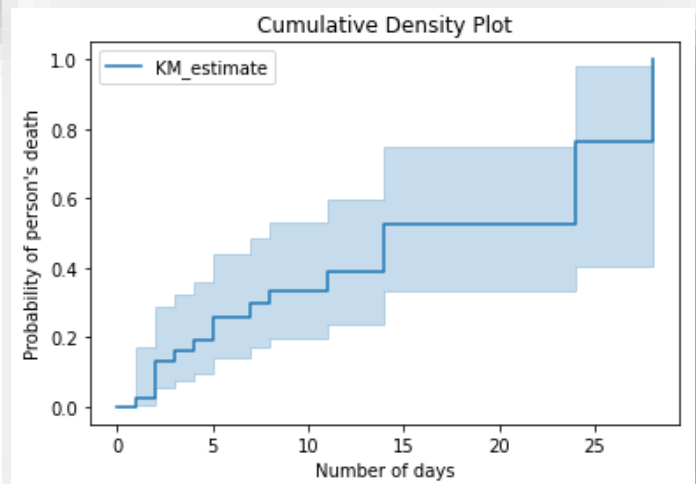
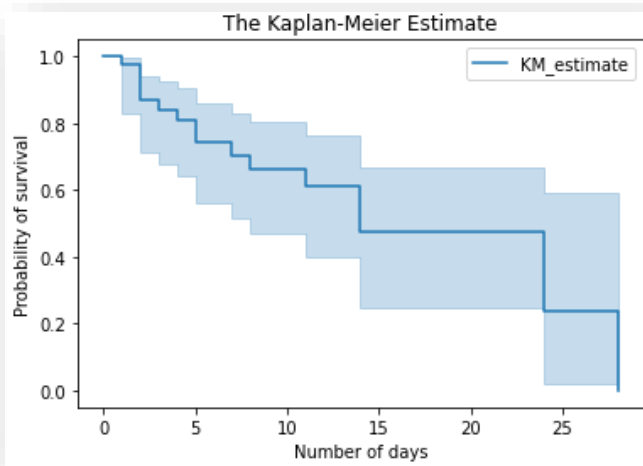
It provides the number of days where on average, 50% of patients survived.

2. Nelson Aalen plots:

It is the probability that an individual (subject) who is under observation at a time 't' has an event (death) at that time. The hazard function $h(t)$ gives us the probability that a subject under observation at time t has an event of interest (death) at that time.

Prediction of the cumulative hazard probability for the complete timeline (NA_estimate):

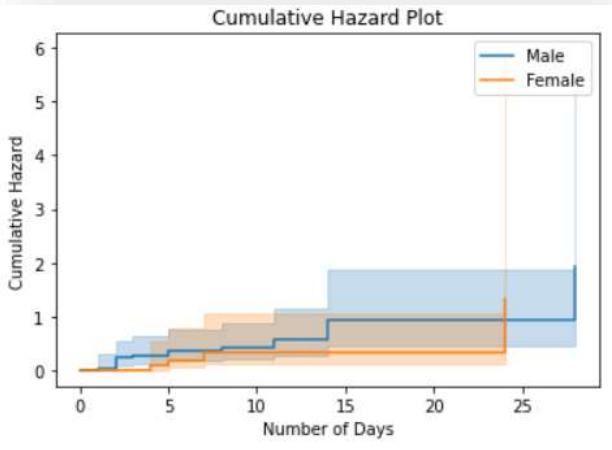
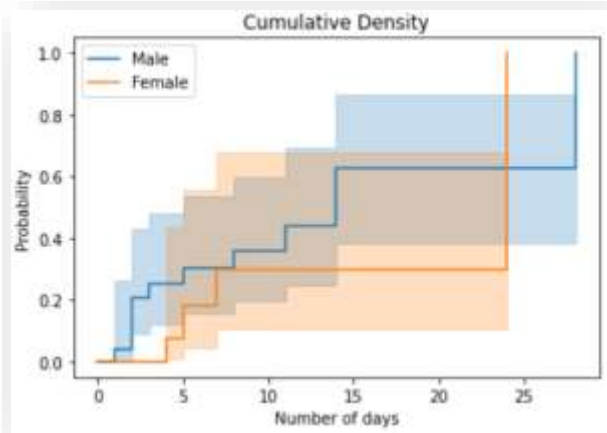
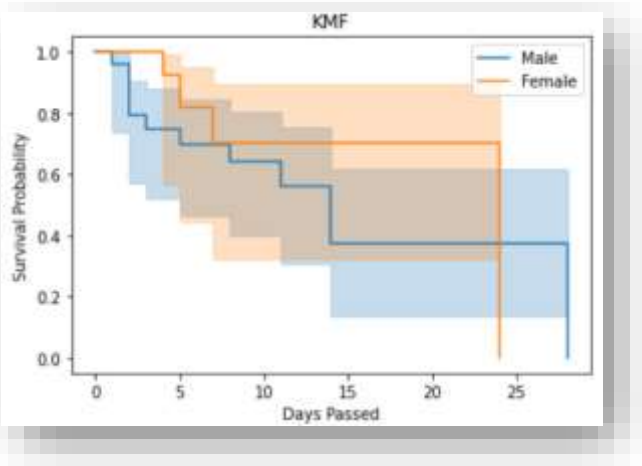
The cumulative hazard has less obvious understanding than the survival functions, but the hazard functions are the basis of more advanced techniques in survival analysis.



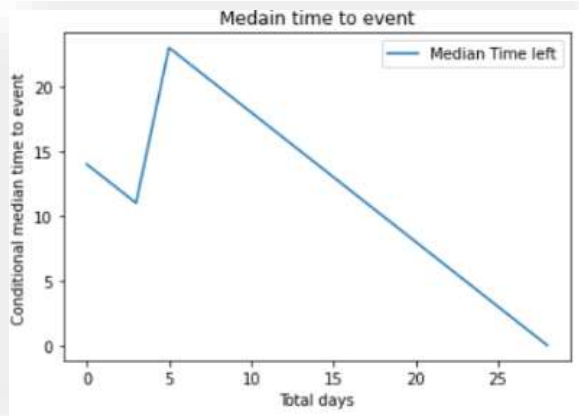
Comparison Table

Timeline	KM_estimate_survival	KM_estimate_death	KM_estimate - Median	NA_estimate
0.0	1.000000	0.000000	14.0	0.000000
1.0	0.973684	0.026316	13.0	0.026316
2.0	0.868421	0.131579	22.0	0.139104
3.0	0.840407	0.159593	21.0	0.171362
4.0	0.810393	0.189607	20.0	0.207076
5.0	0.742860	0.257140	19.0	0.292221
6.0	0.742860	0.257140	18.0	0.292221
7.0	0.703762	0.296238	17.0	0.344853
8.0	0.664664	0.335336	16.0	0.400408
9.0	0.664664	0.335336	15.0	0.400408
10.0	0.664664	0.335336	14.0	0.400408
11.0	0.609276	0.390724	13.0	0.483742
13.0	0.609276	0.390724	11.0	0.483742
14.0	0.473881	0.526119	14.0	0.719853
15.0	0.473881	0.526119	13.0	0.719853
17.0	0.473881	0.526119	11.0	0.719853
19.0	0.473881	0.526119	9.0	0.719853
23.0	0.473881	0.526119	5.0	0.719853
24.0	0.236941	0.763059	4.0	1.219853
28.0	0.000000	1.000000	0.0	2.219853

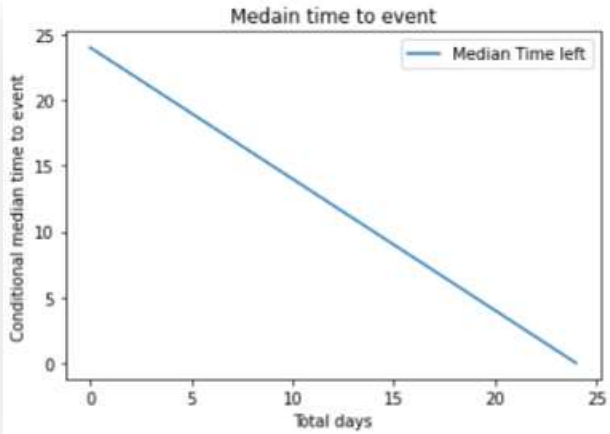
Let's divide our data into 2 groups: Male and Female. Our goal here is to check if there is any significant difference in survival rate if we divide our data set based on sex.



For Male



For Female



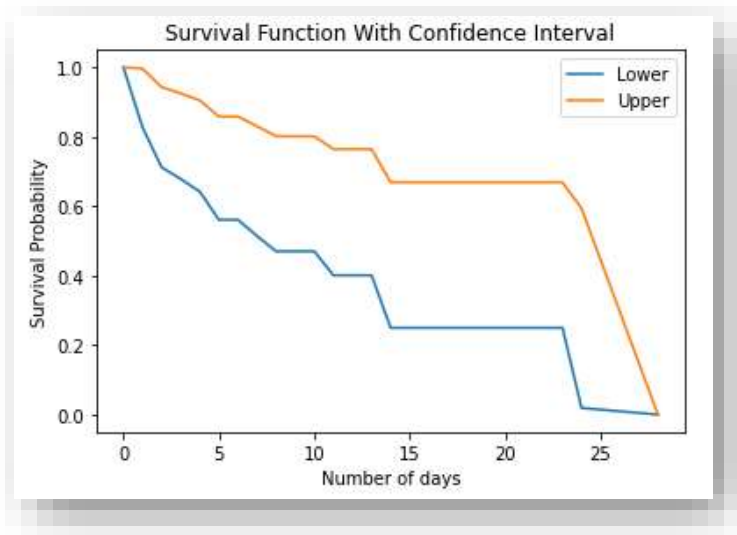
Comparison Table

Timeline	KM_estimate _survival_male	KM_estimate _survival_female	KM_estimate _death_male	KM_estimate _death_female	NA_estimate _male	NA_estimate _female	Male_ median	Female_ median
0.0	1.000000	1.000000	0.000000	0.000000	0.000000	0.000000	14.0	24.0
1.0	0.958333		0.041667		0.041667		13.0	
2.0	0.791667	1.000000	0.208333	0.000000	0.228219	0.000000	12.0	22.0
3.0	0.747685		0.252315		0.283774		11.0	20.0
4.0		0.923077		0.076923		0.076923		
5.0	0.697840	0.820513	0.302160	0.179487	0.350441	0.188034	23.0	19.0
6.0	0.697840		0.302160		0.350441		22.0	
7.0		0.703297		0.296703		0.330891		17.0
8.0	0.639686		0.360314		0.433774		20.0	
9.0	0.639686	0.703297	0.360314	0.296703	0.433774	0.330891	19.0	15.0
10.0	0.639686		0.360314		0.433774		18.0	
11.0	0.559725	0.703297	0.440275	0.296703	0.558774	0.330891	17.0	13.0
13.0	0.559725		0.440275		0.558774		15.0	
14.0	0.373150		0.626850		0.925441		14.0	
15.0	0.373150		0.626850		0.925441		13.0	
17.0	0.373150	0.703297	0.626850	0.296703	0.925441	0.330891	11.0	7.0
19.0	0.373150		0.626850		0.925441		9.0	
23.0		0.703297		0.296703		0.330891		1.0
24.0		0.000000		1.000000		1.330891		0.0
28.0	0.000000		1.000000		1.925441		0.0	

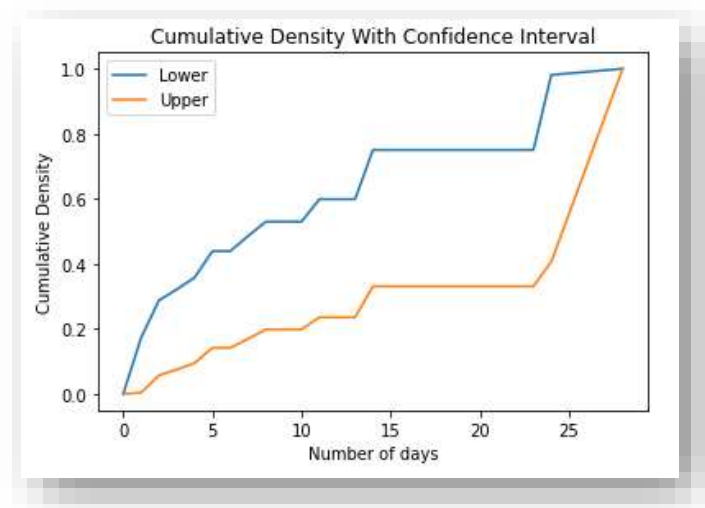
Confidence Interval

A Confidence Interval is a range of values we are fairly sure our true value lies in.

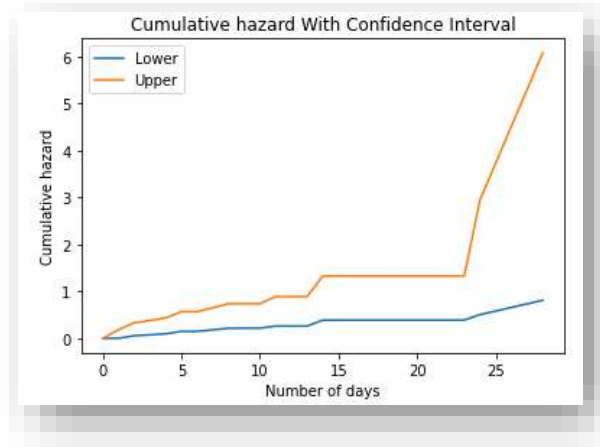
There is a 95% probability that, in the future, the true value of the given function will fall within lower bound and upper bound. The plots for these functions is given below.



The values of the above graph is represented in columns KM_estimate_lower_survival and KM_estimate_upper_survival



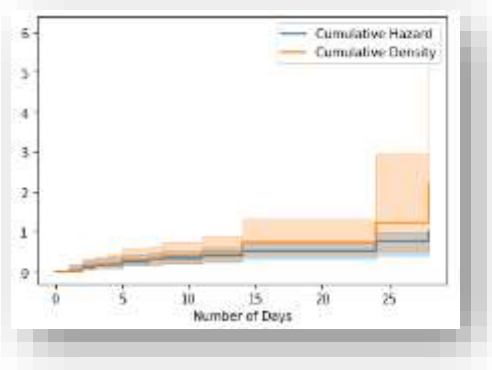
The values of the above graph is represented in columns KM_estimate_lower_death and KM_estimate_upper_death



The values of the above graph is represented in columns NA_estimate_lower and NA_estimate_upper.

Timeline	KM_estimate_lower_survival	KM_estimate_upper_survival	KM_estimate_lower_death	KM_estimate_upper_death	NA_estimate_lower_hazard	NA_estimate_upper_hazard
0.0	1.000000	1.000000	0.000000	0.000000	0.000000	0.000000
1.0	0.827513	0.996251	0.172487	0.003749	0.003707	0.186818
2.0	0.712346	0.943010	0.287654	0.056990	0.057860	0.334428
3.0	0.678581	0.924999	0.321419	0.075001	0.076845	0.382129
4.0	0.642216	0.905009	0.357784	0.094991	0.098327	0.436103
5.0	0.561004	0.858251	0.438996	0.141749	0.150342	0.567993
6.0	0.561004	0.858251	0.438996	0.141749	0.150342	0.567993
7.0	0.514149	0.830664	0.485851	0.169336	0.182260	0.652492
8.0	0.470177	0.801639	0.529823	0.198361	0.216944	0.739022
9.0	0.470177	0.801639	0.529823	0.198361	0.216944	0.739022
10.0	0.470177	0.801639	0.529823	0.198361	0.216944	0.739022
11.0	0.401006	0.764395	0.598994	0.235605	0.263009	0.889726
13.0	0.401006	0.764395	0.598994	0.235605	0.263009	0.889726
14.0	0.249745	0.668977	0.750255	0.331023	0.390194	1.328027
15.0	0.249745	0.668977	0.750255	0.331023	0.390194	1.328027
17.0	0.249745	0.668977	0.750255	0.331023	0.390194	1.328027
19.0	0.249745	0.668977	0.750255	0.331023	0.390194	1.328027
23.0	0.249745	0.668977	0.750255	0.331023	0.390194	1.328027
24.0	0.018837	0.593320	0.981163	0.406680	0.505517	2.943599
28.0	0.000000	0.000000	1.000000	1.000000	0.811008	6.076075

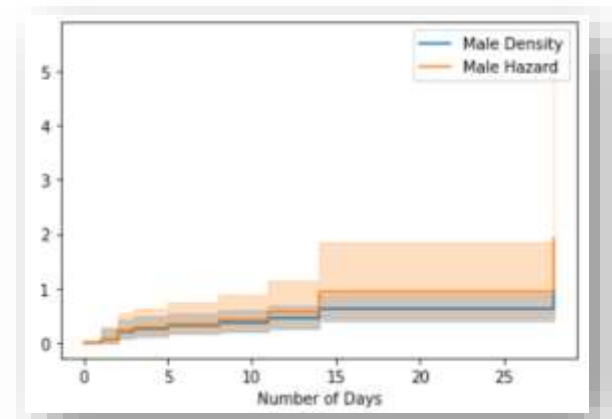
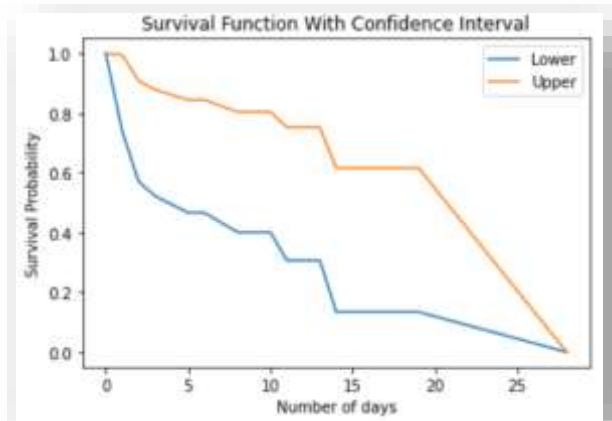
A comparison of cumulative hazard and cumulative density function is given below.



Cumulative Density refers to the death function. The death function is seen to be increasing higher than the hazard function as the number of days' increases.

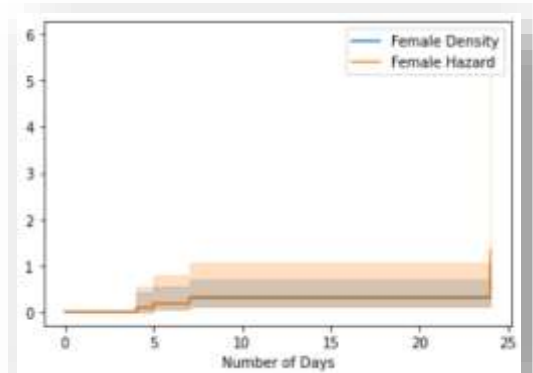
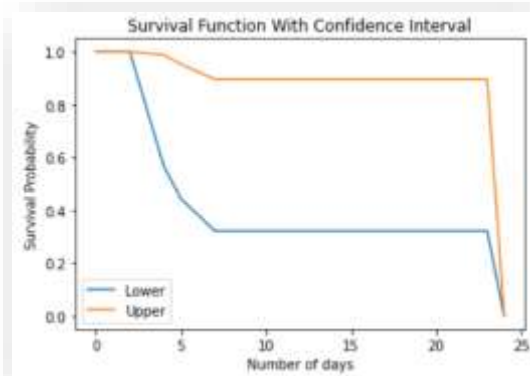
For male population:

Timeline	KM_estimate_ lower_survival	KM_estimate_ upper_survival
0.0	1.000000	1.000000
1.0	0.739206	0.994024
2.0	0.569847	0.907517
3.0	0.522138	0.877998
5.0	0.466610	0.843834
6.0	0.466610	0.843834
8.0	0.401169	0.803690
9.0	0.401169	0.803690
10.0	0.401169	0.803690
11.0	0.306486	0.752191
13.0	0.306486	0.752191
14.0	0.134471	0.616114
15.0	0.134471	0.616114
17.0	0.134471	0.616114
19.0	0.134471	0.616114
28.0	0.000000	0.000000



For female population:

Timeline	KM_estimate_ lower_survival	KM_estimate_ upper_survival
0.0	1.000000	1.000000
2.0	1.000000	1.000000
4.0	0.566359	0.988794
5.0	0.444163	0.952922
7.0	0.321909	0.896464
9.0	0.321909	0.896464
11.0	0.321909	0.896464
17.0	0.321909	0.896464
23.0	0.321909	0.896464
24.0	0.000000	0.000000



3. Log-Rank Test

Goal: To see if there is any significant difference between the groups being compared.

Null Hypothesis: The null hypothesis states that there is no significant difference between the groups being studied. If there is a significant difference between these groups, then we have to reject our null hypothesis.

How do we say that there is a significant difference?

The statistical significance is denoted by a p-value between 0 and 1. The smaller the p-value, the greater the statistical difference between groups being studied. Notice that here our goal is to find if there is any difference between the groups we are comparing. If yes, then we can do more research on why there are lower survival chances for a particular group based on various information like their diet, lifestyle, etc.

Less than (5% = 0.05) P-value means that there is a significant difference between the groups that we compared. We can partition our groups based on their sex, age, race, method of treatment, etc.

The obtained p value for our dataset is **0.348**.

t_0	-1		
null_distribution	chi squared		
degrees_of_freedom	1		
test_name	logrank_test		
test_statistic	p	-log2(p)	
0	0.88	0.35	1.52

Hence, it can be concluded that sex doesn't have an impact on the probability of survival days. However, the obtained results may be used for diseases specific to a certain gender.

4. Cox Proportional Hazard:

The Cox proportional hazard model is basically a regression model generally used by medical researchers to find out the relationship between the survival time of a subject and one or more predictor variables. In short, we want to find out how different parameters like age, sex, weight, height affects the length of survival for a subject.

model	lifelines.CoxPHFitter
duration col	'Time'
event col	'dead'
baseline estimation	breslow
number of observations	38
number of events observed	16
partial log-likelihood	-39.05
time fit was run	2020-11-28 12:21:39 UTC

	Coef	Exp(coef)	Se(coef)	Coef lower 95%	Coef upper 95%	Exp(coef) lower 95%	Exp(coef) upper 95%	Z	P	-log2(p)
Age	-0.05	0.95	0.03	-0.11	0.01	0.90	1.01	-1.49	0.14	2.87
Karnofosky score	-0.09	0.92	0.03	-0.14	-0.04	0.87	0.96	-3.34	<0.005	10.23
Gender	-0.12	0.89	0.66	-1.42	1.18	0.24	3.26	-0.18	0.86	0.22
ACT_mean	-0.01	0.99	0.03	-0.06	0.05	0.94	1.05	-0.23	0.82	0.29
Angle_mean	-0.16	0.85	0.14	-0.44	0.12	0.65	1.13	-1.10	0.27	1.89

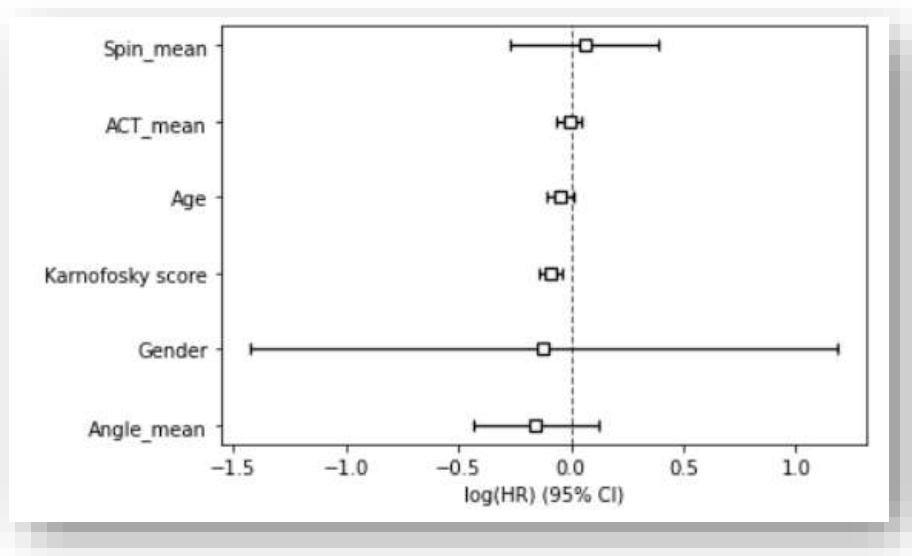
Concordance 0.78

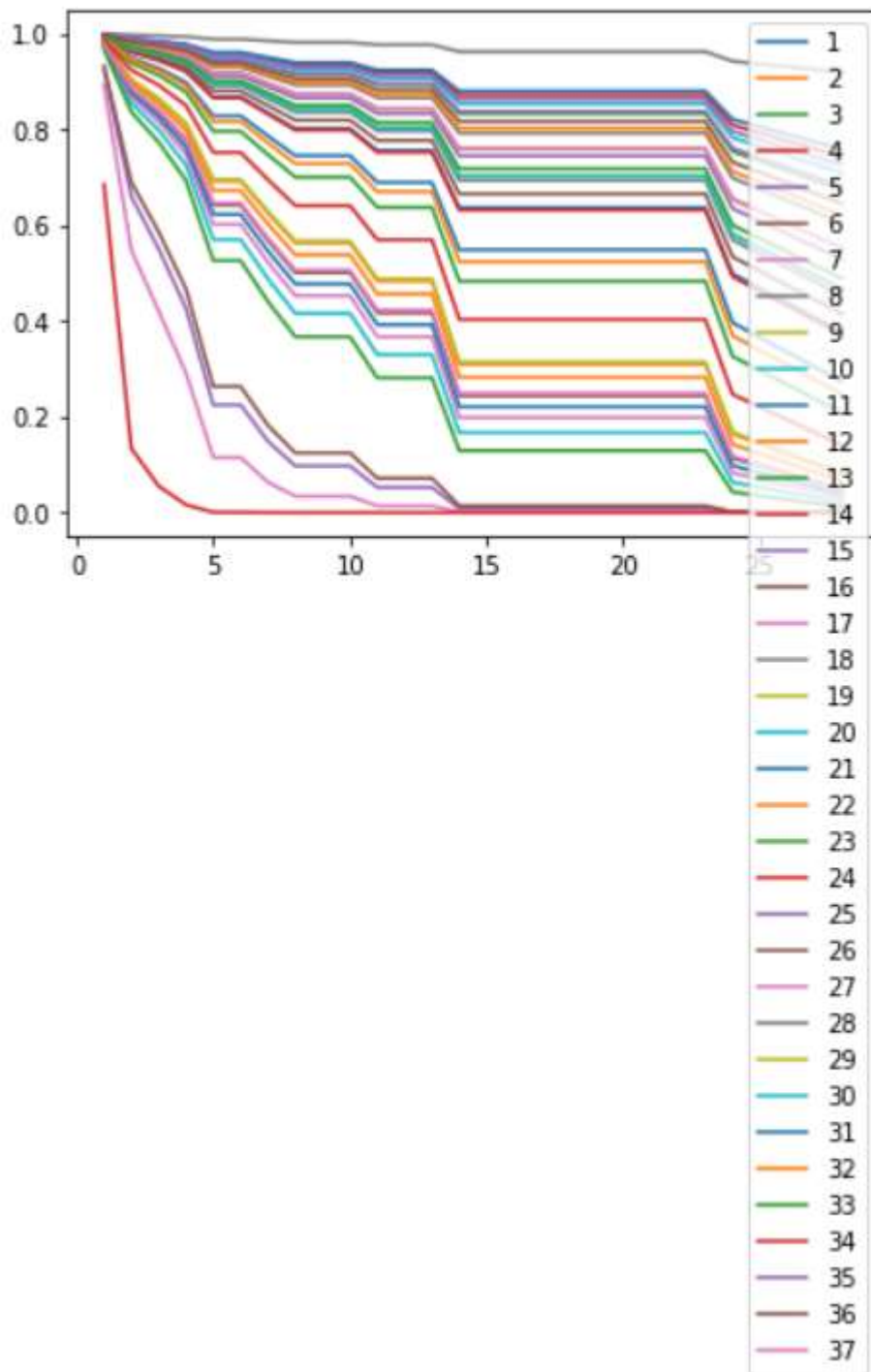
Partial AIC 90.11

log-likelihood ratio test 10.46 on 6 df

-log2(p) of ll-ratio test 3.23

From this it is evident that the **Karnofsky Score** is a significant parameter for determining the survival days.





The above graph depicts the deteriorating nature of the patient's health with respect to time.

MACHINE LEARNING MODEL

Pre-processed Data format:

Index Number	Time	ACT	Angle	Spin	Dead
0	19	168.00	3	12	0.0
1	19	191.00	4	11	0.0
2	19	211.00	3	0	0.0
3	19	87.00	2	3	0.0
4	19	7.19	3	1	0.0

Description of the data with no changes to existing values:

	ACT	Angle	Spin	Dead
Count	736919.000000	736919.000000	736919.000000	736919.000000
Mean	18.440548	7.441863	6.657768	0.324682
Std	80.283599	4.271033	5.429378	0.468256
Min	0.000000	0.000000	0.000000	0.000000
25%	2.560000	4.000000	2.000000	0.000000
50%	5.060000	7.000000	5.000000	0.000000
75%	14.690000	11.000000	12.000000	1.000000

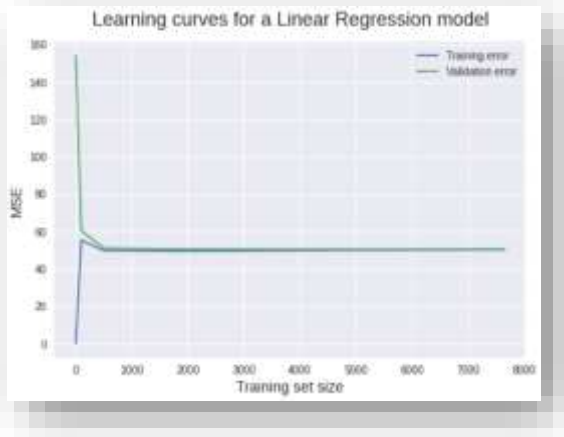
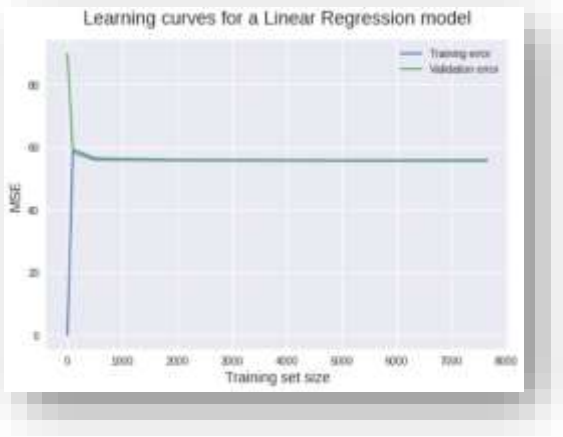
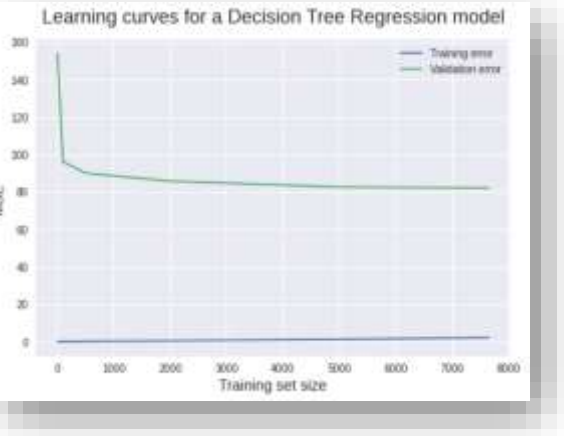
Description of the data with imputation of missing values using KNN:

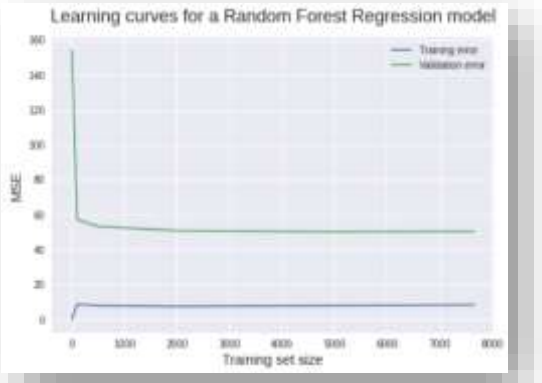
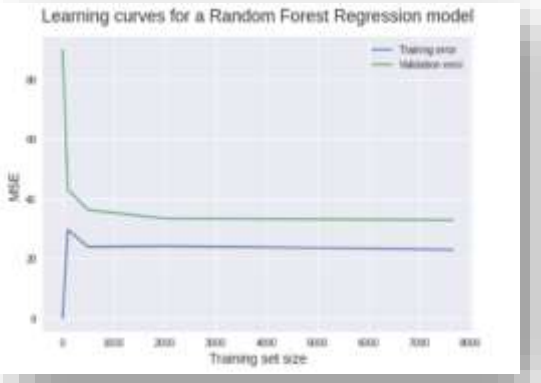
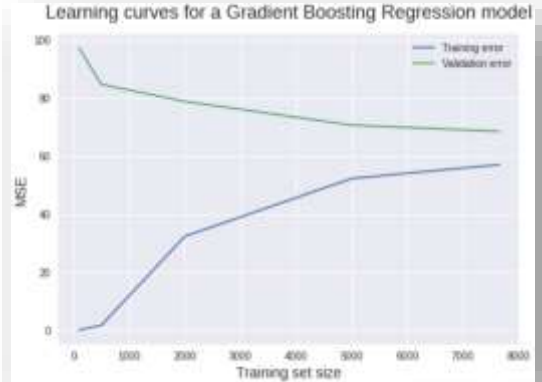
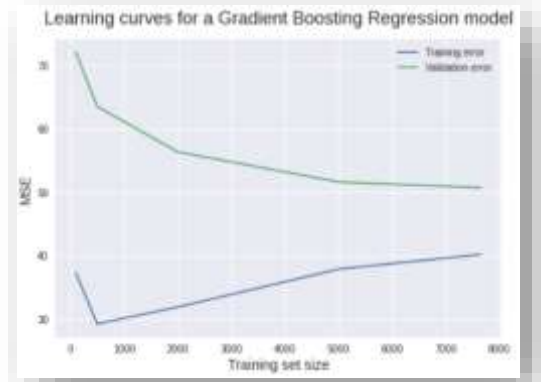
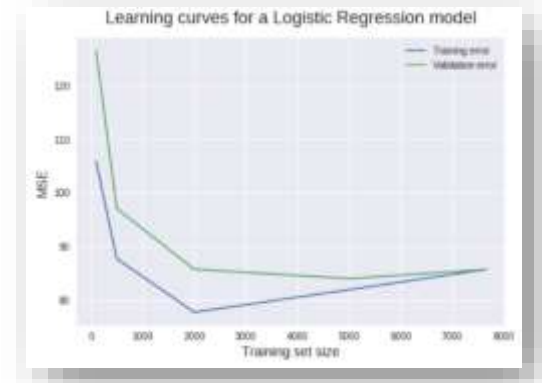
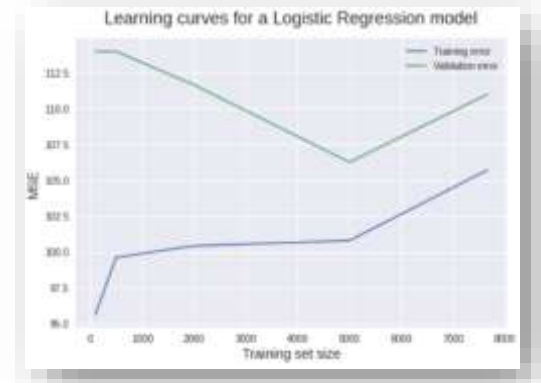
	ACT	Angle	Spin	Dead
Count	2.939771e+06	2.939771e+06	2.939771e+06	2.939771e+06
Mean	1.394875e+01	7.676912e+00	3.579843e+00	3.164388e+00
Std	7.538866e+00	4.142520e+01	3.737610e+00	3.773796e+00
Min	1.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
25%	8.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00
50%	1.400000e+01	3.310000e+00	3.000000e+00	1.964812e+00
75%	1.900000e+01	7.508571e+00	5.354477e+00	4.179399e+00

The various Machine Learning models implemented without filling the missing values (Count approximately equal to 7.3 lakhs) and imputation using the KNN models (Count approximately equal to 29 lakhs). The resulting accuracies (in percentage) have been recorded in the below table:

Index No	Model	Training Accuracy (No change)	Testing Accuracy (No change)	Training Accuracy (Imputation using KNN)	Testing Accuracy (Imputation using KNN)
1	Linear Regression	50.36	50.10	55.57	55.57
2	Decision Tree	25.64	54.99	26.67	33.71
3	Random Forest	27.14	47.22	27.01	31.77
4	Gradient Boosting Classifier	38.38	37.99	66.21	66.26
5	Logistic Regression	24	24	38	38
6	Multi-Layer Perceptron Classifier	40	40	67	67

Learning curves are plotted below for the trained model:

Model	No change dataset	Imputation using KNN
Linear Regression	 Learning curves for a Linear Regression model The plot shows training error (blue line) and validation error (green line) both decreasing and stabilizing at approximately 50 MSE as training set size increases from 0 to 8000.	 Learning curves for a Linear Regression model The plot shows training error (blue line) and validation error (green line) both decreasing and stabilizing at approximately 55 MSE as training set size increases from 0 to 8000.
Decision Tree	 Learning curves for a Decision Tree Regression model The plot shows training error (blue line) remaining near 0 MSE, while validation error (green line) decreases and stabilizes at approximately 85 MSE as training set size increases from 0 to 8000.	 Learning curves for a Decision Tree Regression model The plot shows training error (blue line) stabilizing at approximately 25 MSE, while validation error (green line) decreases and stabilizes at approximately 65 MSE as training set size increases from 0 to 8000.

Random Forest	 Learning curves for a Random Forest Regression model	 Learning curves for a Random Forest Regression model
Gradient Boosting Classifier	 Learning curves for a Gradient Boosting Regression model	 Learning curves for a Gradient Boosting Regression model
Logistic Regression	 Learning curves for a Logistic Regression model	 Learning curves for a Logistic Regression model

Multi-Layer Perceptron Classifier

